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Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

OBJECTIVE1:

CODE:

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

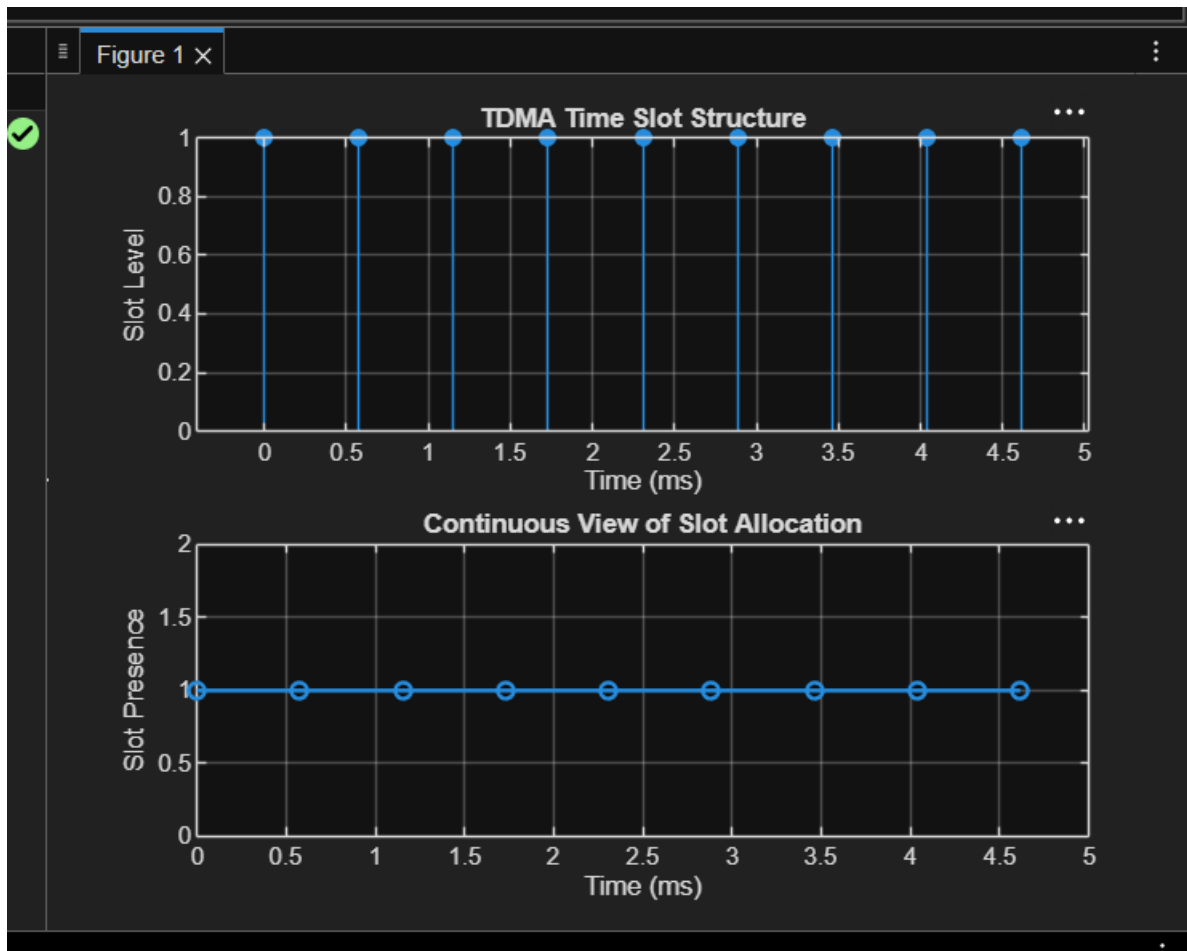
```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```



Objective 2:

Code:

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

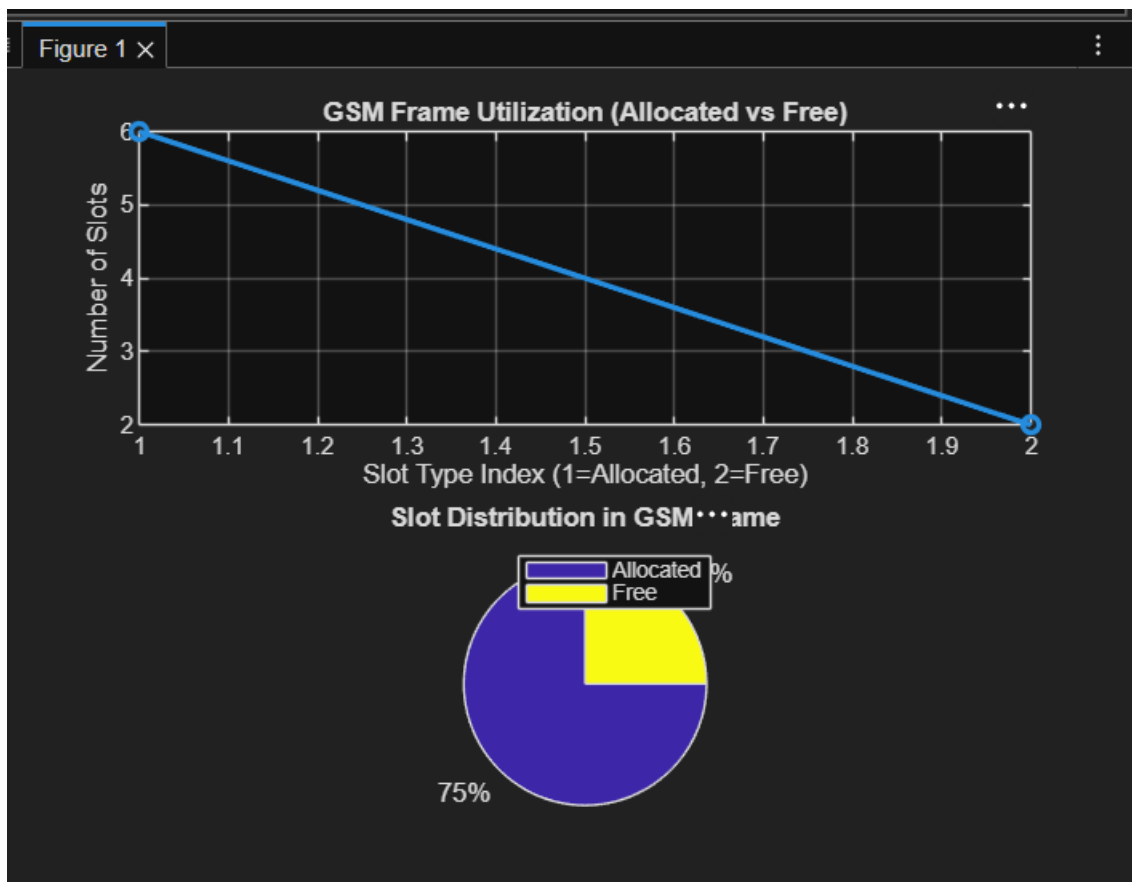
```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```

subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')

```



OBJECTIVE 3:

CODE:

```
clc;
clear;
close all;

Users = 1:8;      % Total users
TotalSlots = 5;   % Available slots
Alloc = zeros(1,8); % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked
disp('User Wise Slot Allocation')
disp(table(Users', Alloc', ...
    'VariableNames', {'User','Status_Allocated'}))

figure
% Graph 1: Allocation per user
subplot(2,1,1)
stem(Users, Alloc, 'filled', 'LineWidth', 2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

% Graph 2: System view
subplot(2,1,2)
plot(Users, Alloc, '-o', 'LineWidth', 2)
```

hold on

```
plot(Users, Blocked, '-s', 'LineWidth', 2)
```

```
title('TDMA Slot Utilization Analysis')
```

```
xlabel('User Index')
```

```
ylabel('Status')
```

```
legend('Allocated','Blocked')
```

grid on

